PREDICTIVE ANALYTICS

IBM SPSS MODELER PRACTICAL FILE

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ABSTRACT

Data is often collected in an order that is not immediately useful for analysis. A dataset may contain customer names, regions, product categories, sales values, ratings, and quantities in a random or entry-based order. Sorting arranges records in a meaningful sequence so that users can identify the highest or lowest values, locate records more easily, and prepare clear reports.

This project demonstrates sorting in **IBM SPSS Modeler** by using its **Sort node**. A fictional sales dataset containing 20 records is used. The same records are sorted by Sales in ascending order (lowest to highest) and descending order (highest to lowest). A multi-field example is also included, where records are sorted first by Region and then by Customer.

The report covers the project objectives, software overview, dataset design, field definitions, workflow, node configuration, execution, expected results, validation checks, common errors, applications, limitations, and conclusion. It highlights that sorting changes record sequence but does not calculate new values or remove records.

The practical outcome is a repeatable workflow: import data, inspect fields, add a Sort node, select the sorting field, choose ascending or descending order, run the stream, and verify the output.

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1. INTRODUCTION

Organisations collect large amounts of data from sales transactions, customer interactions, inventory systems, surveys, and many other sources. Before the data can be studied, it is often useful to arrange records in a consistent order. Sorting is one of the simplest but most important data preparation operations.

For example, a sales manager may want to view the smallest sales values first to identify low-performing transactions. The same manager may want the largest sales values first to identify top transactions. Sorting by region and then customer name can also make a report easier to read.

IBM SPSS Modeler is a visual data-mining and analytics application. Users build a stream by connecting nodes. Each node performs a task such as reading data, selecting fields, deriving values, sorting records, modelling, or exporting results. The Sort node is used to arrange records according to one or more selected fields.

This project focuses on the practical use of the Sort node. It does not attempt to build a predictive model. Instead, it demonstrates a foundational data-preparation step that can be used before reporting, grouping, modelling, or exporting data.

2. OBJECTIVES AND SCOPE

Main objectives

1. To understand the purpose of sorting in data preparation.
2. To become familiar with the IBM SPSS Modeler stream environment.
3. To create or import a structured dataset with numeric and categorical fields.
4. To configure the Sort node for ascending order.
5. To configure the Sort node for descending order.
6. To demonstrate sorting by more than one field.
7. To compare the input and sorted output and validate the result.

Scope of the project

The project uses a small fictional sales dataset with 20 records. The main sorting field is Sales (■), a numeric field. Ascending order means the smallest sales amount appears first. Descending order means the largest sales amount appears first. A secondary example sorts by Region and Customer.

Out of scope

The project does not cover advanced statistical tests, predictive modelling, database administration, or automated production deployment. The exact screen layout and labels may vary slightly by IBM SPSS Modeler version.

Learning outcome: After completing the steps, a student should be able to create a simple stream and use the Sort node to produce correctly ordered records.

3. IBM SPSS MODELER OVERVIEW

IBM SPSS Modeler is a visual analytics environment used to prepare data, explore patterns, and build analytical workflows. Rather than writing every operation as code, users connect graphical nodes on a stream canvas. Data moves from one node to the next when the stream is executed.

Common node categories

- **Sources:** Read data from files, databases, and other supported locations.
- **Record operations:** Select, sample, aggregate, or sort records.
- **Field operations:** Derive, restructure, or select variables.
- **Graphs and outputs:** Inspect distributions, tables, and results.
- **Modelling:** Build models for prediction or segmentation.

The Sort node

The Sort node is a record-operation node. It orders records based on one or more fields selected by the user. For each field, the user specifies whether the order should be ascending or descending. When multiple fields are selected, their sequence determines the sort priority.

For example, if Region is the first sort key and Customer is the second, records are grouped alphabetically by Region, and customers within each region are arranged alphabetically. The Sort node does not create a new measure; it changes the sequence in which records are presented downstream.

4. UNDERSTANDING DATA SORTING

Ascending order

Ascending order arranges values from lower to higher. For numbers, 100 comes before 250 and 900. For text, values are generally arranged alphabetically according to the software's collation rules. For dates, ascending order usually means oldest to newest.

Descending order

Descending order arranges values from higher to lower. For numbers, 900 comes before 250 and 100. For text, values are generally arranged in reverse alphabetical order. For dates, descending order usually means newest to oldest.

Single-field and multi-field sorting

A single-field sort uses one key, such as Sales. A multi-field sort uses two or more keys. The first key has the highest priority. The next key is used to order records that have equal values in the earlier key.

Sort key	Direction	Purpose
Sales	Ascending	Find the smallest sales amounts first
Sales	Descending	Find the largest sales amounts first
Region, Customer	Ascending, Ascending	Group regions and alphabetise names within each region

A good sort operation requires correct field types. A numeric amount should be stored as a number, not as text, otherwise the ordering may be lexical rather than numeric.

5. DATASET DESCRIPTION

A fictional sales dataset has been designed for this project. It represents 20 customer transactions. Each row is one record, and each column is one field. The dataset is intentionally small so that the output can be checked manually.

Field	Type	Description
ID	Integer	Unique record identifier from 101 to 120
Customer	String	Fictional customer first name
Region	String	Sales region: North, South, East, or West
Category	String	Product category
Sales (■)	Integer	Sales amount in Indian rupees
Rating	Decimal	Illustrative customer rating
Units	Integer	Number of units in the transaction

Data quality considerations

Before sorting, check that the column names are clear, each field has the correct measurement type, sales values do not contain currency symbols stored as text, and there are no unintended blank rows. The ID field should be unique so that individual records can be traced after sorting.

The sample contains no missing values. In a real project, missing values, duplicate IDs, inconsistent region spellings, and invalid numeric values should be reviewed before the sort operation.

6. SAMPLE DATA (INPUT)

The following table shows all 20 fictional records before sorting. The input order is based on the record ID and is not arranged by sales amount.

ID	Customer	Region	Category	Sales (■)	Rating	Units
101	Aarav	North	Electronics	12500	4.2	12
102	Diya	West	Furniture	8200	3.8	8
103	Kabir	South	Electronics	15600	4.6	15
104	Meera	East	Stationery	2400	4.1	5
105	Ishaan	North	Furniture	9100	3.5	9
106	Anaya	West	Stationery	1800	4.4	4
107	Vivaan	South	Electronics	13200	4.0	11
108	Sara	East	Furniture	7600	4.3	7
109	Reyansh	North	Stationery	3200	3.9	6
110	Aditi	West	Electronics	14800	4.7	14
111	Arjun	South	Furniture	9900	3.7	10
112	Myra	East	Stationery	2100	4.5	5
113	Dev	North	Electronics	11900	4.1	12
114	Kiara	West	Furniture	8700	4.0	8
115	Rohan	South	Stationery	2900	3.6	6
116	Ira	East	Electronics	16200	4.8	16
117	Advik	North	Furniture	10400	4.2	10
118	Tara	West	Stationery	2600	4.3	5
119	Yash	South	Electronics	13700	4.4	13
120	Nisha	East	Furniture	9300	3.9	9

Table 1: Illustrative input dataset. All names and values are fictional and used only for demonstration.

The same input table will be used for both ascending and descending examples. This makes it easier to compare the results and confirm that sorting has changed only the row sequence.

7. PROJECT METHODOLOGY

The project follows a simple data-preparation workflow. First, the input data is created in a spreadsheet or CSV file. Next, the file is brought into IBM SPSS Modeler through an appropriate source node. The data is inspected to confirm that fields have been read correctly. A Sort node is then connected and configured.

Step	Activity	Expected output
1	Prepare CSV / spreadsheet	Structured input file
2	Create a source node	Data is available in the stream
3	Inspect field metadata	Correct field names and types
4	Add a Sort node	Sort operation connected
5	Select sort field and direction	Ascending or descending rule set
6	Run the stream	Sorted records produced
7	Inspect and validate output	Correct order confirmed

Stream design

Input file → Source node → Sort node → Table / output node

The Table node or another suitable output node can be used to inspect the records after sorting. The stream can be duplicated or the Sort node configuration can be changed to produce the opposite direction.

8. BUILDING THE STREAM IN SPSS MODELER

1. Open IBM SPSS Modeler and create a new stream.
2. Prepare the sample data in a CSV file with the column headings shown in this report.
3. From the Sources palette, select the source node appropriate for the file type (for example, a Var. File node for a delimited text file, depending on the installed version).
4. Place the source node on the stream canvas and configure the file path.
5. Read the file and inspect the preview to confirm that the rows and columns are separated correctly.
6. Check field metadata. Set ID, Sales, and Units as numeric fields; set Customer, Region, and Category as string fields; and set Rating as a numeric decimal field.
7. Add a Sort node from the record-operation palette and connect the source node to it.
8. Add a Table node after the Sort node to inspect the output.

Illustrative stream layout

Node	Role
Source	Reads the sample CSV file
Sort	Orders records using selected fields
Table	Displays the sorted records for checking

Important: The exact source node name depends on the input format and SPSS Modeler release. The key requirement is that the source node supplies correctly typed fields to the Sort node.

9. ASCENDING SORT PROCEDURE

The first practical task is to arrange the records from the lowest sales amount to the highest sales amount.

Configuration steps

1. Open the Sort node settings by double-clicking the node.
2. Locate the list of available fields and the selected sort keys.
3. Move or add **Sales** to the selected sort-field list.
4. Set the order for Sales to **Ascending** (lowest to highest).
5. If the interface offers a separate order column or direction selector, verify that it displays Ascending for Sales.
6. Apply the settings and close the dialog.
7. Run the Sort node or the downstream Table node to execute the stream.
8. Inspect the first and last rows to verify the result.

Expected first and last values

Position	Customer	Sales (■)
First / minimum	Anaya	1,800
Second	Myra	2,100
Third	Meera	2,400
Last / maximum	Ira	16,200

The first record should have the lowest sales value, ■1,800, and the last record should have the highest sales value, ■16,200. If this is not the case, recheck the selected field and direction.

10. DESCENDING SORT PROCEDURE

The second practical task is to arrange the records from the highest sales amount to the lowest sales amount.

Configuration steps

1. Open the Sort node settings. If you want to preserve the ascending result, duplicate the Sort node or save a separate stream version before changing the setting.
2. Select **Sales** as the sort field.
3. Change its direction to **Descending** (highest to lowest).
4. Apply the configuration.
5. Run the stream and open the downstream Table output.
6. Confirm that the first row contains the maximum sales value and the last row contains the minimum sales value.

Expected first and last values

Position	Customer	Sales (■)
First / maximum	Ira	16,200
Second	Kabir	15,600
Third	Aditi	14,800
Last / minimum	Anaya	1,800

Descending order is useful when the purpose is to identify the largest transactions, top customers, highest expenses, or latest dates first. The values remain the same; only their sequence changes.

11. RESULTS AND INTERPRETATION

Ascending result: lowest to highest

ID	Customer	Region	Category	Sales (■)	Rating	Units
106	Anaya	West	Stationery	1800	4.4	4
112	Myra	East	Stationery	2100	4.5	5
104	Meera	East	Stationery	2400	4.1	5
118	Tara	West	Stationery	2600	4.3	5
115	Rohan	South	Stationery	2900	3.6	6
109	Reyansh	North	Stationery	3200	3.9	6
108	Sara	East	Furniture	7600	4.3	7
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119	Yash	South	Electronics	13700	4.4	13
110	Aditi	West	Electronics	14800	4.7	14
103	Kabir	South	Electronics	15600	4.6	15
116	Ira	East	Electronics	16200	4.8	16

Descending result: highest to lowest

ID	Customer	Region	Category	Sales (■)	Rating	Units
116	Ira	East	Electronics	16200	4.8	16
103	Kabir	South	Electronics	15600	4.6	15
110	Aditi	West	Electronics	14800	4.7	14
119	Yash	South	Electronics	13700	4.4	13
107	Vivaan	South	Electronics	13200	4.0	11
101	Aarav	North	Electronics	12500	4.2	12
113	Dev	North	Electronics	11900	4.1	12
117	Advik	North	Furniture	10400	4.2	10
111	Arjun	South	Furniture	9900	3.7	10
120	Nisha	East	Furniture	9300	3.9	9
105	Ishaan	North	Furniture	9100	3.5	9
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102	Diya	West	Furniture	8200	3.8	8
108	Sara	East	Furniture	7600	4.3	7
109	Reyansh	North	Stationery	3200	3.9	6
115	Rohan	South	Stationery	2900	3.6	6
118	Tara	West	Stationery	2600	4.3	5
104	Meera	East	Stationery	2400	4.1	5
112	Myra	East	Stationery	2100	4.5	5
106	Anaya	West	Stationery	1800	4.4	4

The two outputs contain the same 20 records and the same field values. Only the row sequence changes. The ascending output begins at ■1,800, while the descending output begins at ■16,200. This confirms the expected behaviour of the Sort node.

12. VALIDATION AND TROUBLESHOOTING

Validation checklist

- Confirm that the output contains 20 records, the same as the input.
- Confirm that every ID from 101 to 120 appears exactly once.
- For ascending sales, check that each value is less than or equal to the next value.
- For descending sales, check that each value is greater than or equal to the next value.
- Confirm that the minimum and maximum values appear at the expected ends.
- Check that customer names and other fields remain attached to the correct ID.

Common problems and solutions

Problem	Possible reason	Solution
Numbers sort unexpectedly	Sales imported as text	Change field storage/type to numeric and reread data
Wrong direction	Ascending/descending selected incorrectly	Reopen Sort node and verify direction
Some records missing	Filter or selection node upstream	Review earlier nodes and input row count
Ties appear in varying order	No secondary sort key defined	Add ID or another field as a second key
Field not available	Metadata not refreshed or wrong input	Refresh fields / inspect source metadata

When two records have the same sales amount, their relative order may not be meaningful unless a secondary sort key is specified. For a repeatable result, add ID as a secondary key.

13. APPLICATIONS, BENEFITS AND LIMITATIONS

Practical applications

Sorting is useful in sales analysis (highest and lowest sales), finance (largest expenses or balances), education (marks from highest to lowest), inventory (items by stock quantity), human resources (employee records by joining date), and survey analysis (responses by category or score).

Benefits

- Makes large datasets easier to read and inspect.
- Helps identify minimum and maximum values quickly.
- Supports preparation of clear tables and reports.
- Can use multiple fields to create structured output.
- Fits into a repeatable visual workflow in SPSS Modeler.

Limitations and precautions

- Sorting does not explain why a value is high or low.
- Sorting does not remove duplicates or correct inaccurate values.
- Incorrect field types can produce incorrect ordering.
- Very large datasets may require attention to system resources and workflow design.
- Sorting alone is not a statistical analysis or predictive model.

Therefore, sorting should be considered one step in a broader data-preparation process. Data quality, field definitions, and the purpose of the analysis should be checked before drawing conclusions.

14. CONCLUSION AND REFERENCES

Conclusion

This project demonstrated how to sort data in ascending and descending order using the Sort node in IBM SPSS Modeler. A fictional dataset of 20 sales records was prepared and sorted using the Sales field. In ascending order, the smallest sales value appeared first and the largest appeared last. In descending order, the largest sales value appeared first and the smallest appeared last.

The project also explained how to sort by multiple fields, how to validate the output, and how to troubleshoot common issues such as incorrect field types or wrong sort direction. The exercise shows that sorting is a simple but valuable data-preparation operation. It improves readability and helps users find records of interest without changing the underlying data values.

The main learning is that a correct sort depends on selecting the right field, confirming its data type, choosing the required direction, running the stream, and checking the output.

References

1. IBM Documentation, *IBM SPSS Modeler* product documentation and node reference materials, available through IBM Documentation: https://www.ibm.com/docs/
2. IBM SPSS Modeler user interface and installed product help (version-specific guidance).
3. Project dataset: fictional sample data created specifically for this report.

Appendix: suggested screenshots to add before submission

To make the report reflect your own practical work, insert screenshots of: (a) the input data preview, (b) the stream showing Source → Sort → Table, (c) the Sort node configured for ascending Sales, (d) the ascending output, (e) the Sort node configured for descending Sales, and (f) the descending output.

End of Report